Università di Pisa
Department of Physics
Master's Degree in Physics

# Anomalous Transport in Soaring Flights

## Collective Strategies and Intermittent Search Models

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# Contents

**1 Introduction** **1**

**2 The dataset** **2**
2.1 Source . . . 2
2.2 Tracks and metadata . . . 2
2.3 Glider categories . . . 2

**3 Trajectory pre-processing** **6**
3.1 Reading the fixes . . . 6
3.2 Choice of altitude channel . . . 7
3.3 Clock . . . 8
3.4 Repeated positions . . . 8
3.5 Fast tracks . . . 8
3.6 Horizontal excursions . . . 10
3.7 Vertical excursions . . . 13
3.8 Local coordinates and vertical datum . . . 15

# Chapter 1

# Introduction

# Chapter 2

# The dataset

## 2.1 Source

The flights come from the *Coupe Fédérale de Distance* (CFD), the cross-country competition of the French free-flight federation (FFVL). Pilots declare their flights online, and each entry carries the flight's metadata and, in most cases, its GPS track. Paragliders and hang gliders have separate archives.

A season runs from 1 September to 31 August. We use the paraglider seasons from 1999–2000 and the hang-glider seasons from 2001–2002, up to 2025–2026. For each season we download the list of declared flights, kept verbatim as provenance, and the tracks it links. A saved list is never replaced, so each one is a snapshot of the day it was downloaded: the 2025–2026 lists were saved around the end of June 2026 and miss the rest of that season.

## 2.2 Tracks and metadata

Tracks are IGC files [1], plain-text records of the flight. Each `B` record is one *fix*: UTC time, latitude and longitude, a validity flag, and two altitudes, the pressure altitude from the barometer and the GNSS altitude from the satellite fix. The sampling interval depends on the recorder and need not be constant within a flight. We take as the sampling interval of a track the most frequent positive time between consecutive fixes. Most paraglider tracks have an interval of one second, while hang-glider tracks are slower, often 5 or 10 seconds (Figure 2.1). Some recorders log no pressure altitude.

The metadata give the date, the pilot, the take-off and landing sites, the wing model and class, the scored distance and the scoring category of the declared route (free distance, out-and-return, flat or FAI triangle and a few others). The category describes the declared route only. A few paraglider entries have an incomplete date, such as `0000-00-00`, which excludes them from any comparison by date.

## 2.3 Glider categories

Wing design changes speed, glide performance and turning geometry, so paragliders and hang gliders are analysed separately [2]. A paraglider is a flexible canopy while a hang glider has a framed wing, flexible or rigid.

The wing class follows a different system in each discipline. Paraglider classes are those of the EN 926-2 standard, A to D, written with their equivalent in the older LTF scale ("C ou 2"

**Table 2.1.** Paraglider flights per season.

| Season | Flights | With track | Downloaded | Downloaded (% of tracks) |
|---|---|---|---|---|
| 1999–2000 | 10 | 10 | 10 | 100.0 |
| 2000–2001 | 832 | 1 | 1 | 100.0 |
| 2001–2002 | 1,150 | 3 | 3 | 100.0 |
| 2002–2003 | 1,996 | 123 | 123 | 100.0 |
| 2003–2004 | 1,470 | 337 | 337 | 100.0 |
| 2004–2005 | 1,765 | 574 | 574 | 100.0 |
| 2005–2006 | 2,162 | 894 | 894 | 100.0 |
| 2006–2007 | 2,386 | 1,016 | 1,016 | 100.0 |
| 2007–2008 | 3,081 | 1,667 | 1,667 | 100.0 |
| 2008–2009 | 3,974 | 2,614 | 2,613 | 100.0 |
| 2009–2010 | 4,171 | 3,059 | 3,059 | 100.0 |
| 2010–2011 | 4,872 | 4,185 | 4,169 | 99.6 |
| 2011–2012 | 6,744 | 6,075 | 6,065 | 99.8 |
| 2012–2013 | 7,873 | 7,106 | 7,075 | 99.6 |
| 2013–2014 | 9,804 | 9,157 | 9,127 | 99.7 |
| 2014–2015 | 13,793 | 13,293 | 13,254 | 99.7 |
| 2015–2016 | 13,982 | 13,703 | 13,696 | 99.9 |
| 2016–2017 | 15,760 | 15,534 | 15,528 | 100.0 |
| 2017–2018 | 14,917 | 14,682 | 14,676 | 100.0 |
| 2018–2019 | 16,702 | 16,517 | 16,508 | 99.9 |
| 2019–2020 | 10,966 | 10,868 | 10,863 | 100.0 |
| 2020–2021 | 9,500 | 9,483 | 9,480 | 100.0 |
| 2021–2022 | 14,552 | 14,507 | 14,502 | 100.0 |
| 2022–2023 | 11,590 | 11,561 | 11,560 | 100.0 |
| 2023–2024 | 10,372 | 10,362 | 10,360 | 100.0 |
| 2024–2025 | 10,906 | 10,885 | 10,884 | 100.0 |
| 2025–2026 | 8,013 | 8,009 | 8,008 | 100.0 |
| Total | 203,343 | 186,225 | 186,052 | 99.9 |

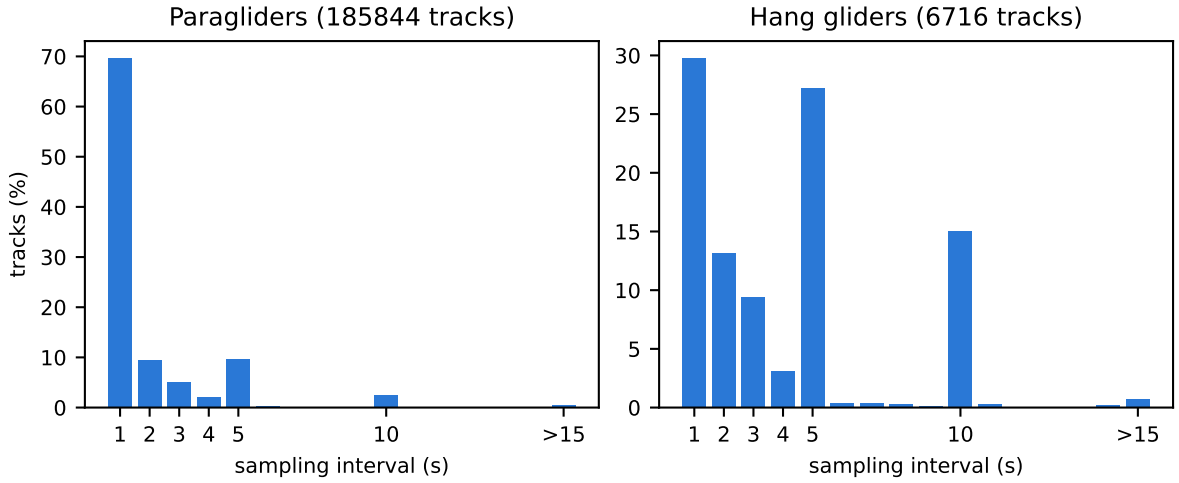



**Figure 2.1.** Sampling interval of the tracks, the most frequent positive time between consecutive fixes. One bar per second; the last bar holds every interval longer than 15 s.

**Table 2.2.** Hang-glider flights per season.

| Season | Flights | With track | Downloaded | Downloaded (% of tracks) |
|---|---|---|---|---|
| 2001–2002 | 248 | 0 | 0 | – |
| 2002–2003 | 281 | 6 | 6 | 100.0 |
| 2003–2004 | 256 | 28 | 25 | 89.3 |
| 2004–2005 | 244 | 35 | 25 | 71.4 |
| 2005–2006 | 365 | 80 | 76 | 95.0 |
| 2006–2007 | 337 | 100 | 96 | 96.0 |
| 2007–2008 | 388 | 136 | 132 | 97.1 |
| 2008–2009 | 344 | 155 | 155 | 100.0 |
| 2009–2010 | 324 | 136 | 136 | 100.0 |
| 2010–2011 | 425 | 147 | 147 | 100.0 |
| 2011–2012 | 117 | 103 | 103 | 100.0 |
| 2012–2013 | 428 | 406 | 406 | 100.0 |
| 2013–2014 | 423 | 409 | 409 | 100.0 |
| 2014–2015 | 628 | 595 | 591 | 99.3 |
| 2015–2016 | 417 | 413 | 413 | 100.0 |
| 2016–2017 | 468 | 462 | 462 | 100.0 |
| 2017–2018 | 342 | 331 | 331 | 100.0 |
| 2018–2019 | 604 | 594 | 594 | 100.0 |
| 2019–2020 | 378 | 373 | 373 | 100.0 |
| 2020–2021 | 328 | 328 | 328 | 100.0 |
| 2021–2022 | 570 | 568 | 568 | 100.0 |
| 2022–2023 | 376 | 374 | 374 | 100.0 |
| 2023–2024 | 437 | 437 | 436 | 99.8 |
| 2024–2025 | 353 | 352 | 352 | 100.0 |
| 2025–2026 | 178 | 178 | 178 | 100.0 |
| Total | 9,259 | 6,746 | 6,716 | 99.6 |

reads “C or 2”), plus the CIVL Competition Class and a few other labels, such as tandem or uncertified wings. EN classes rank passive safety and the skill needed to recover from a collapse, and say nothing direct about speed or glide ratio [3]. Hang-glider classes are those of the FAI [4]: Class 1, with its Sport sub-class, for wings controlled by weight shift alone, and Classes 2 and 5 for rigid wings with movable control surfaces.

# Chapter 3

# Trajectory pre-processing

The pre-processing turns each IGC file into a trajectory fit for analysis. This chapter describes its steps in the order they run, one section per step. Some steps delete single fixes, and the two excursion rules also split the flight; others judge the whole flight and keep or drop it. Each step answers one question about the recording:

1. **Reading the fixes** (Section 3.1). Each record of the file becomes a fix with a time, a 2D position and two altitudes, one from the barometer and one from the GNSS receiver. Records that cannot be read are discarded.
2. **Choice of altitude channel** (Section 3.2). The analysis uses the GNSS altitude, so a flight whose GNSS altitude is mostly missing or never moves is dropped, and in the flights kept the few fixes without it are deleted. The barometer presence is checked too because, when available, it serves as an independent witness of real vertical motion.
3. **Clock** (Section 3.3). Time must run forward: a fix that does not come after every fix before it is deleted.
4. **Repeated positions** (Section 3.4). A recorder that stops updating repeats its last position; the copies are deleted.
5. **Fast tracks** (Section 3.5). A track whose typical speed is too high is not a free flight, and is dropped.
6. **Horizontal excursions** (Section 3.6). A fix the glider could not have reached at its top speed is a recorder error: it is deleted if the track comes back soon after, and otherwise the flight is split where the track jumps.
7. **Vertical excursions** (Section 3.7). The same rule on the GNSS altitude, with a top vertical speed read off the barometer.

What comes out is a set of segments, pieces of the flight that no rule split, with a time, a position and an altitude on every fix. Table 3.1 sums up the steps and their thresholds.

## 3.1 Reading the fixes

Each `B` record gives a fix $(t_i, \varphi_i, \lambda_i)$, with latitude and longitude in degrees, together with the validity flag and the two altitudes $h_i^{\text{baro}}$ and $h_i^{\text{GNSS}}$. A record that does not follow the fixed layout of the format, or whose time of day or position is impossible, is discarded. A missing or

**Table 3.1.** The steps of the pre-processing, in the order they run. Where two values are given, the first holds for paragliders and the second for hang gliders; the two excursion rules share the window $W$.

| Section | Judges | Removes | Thresholds |
|---|---|---|---|
| 3.1 Reading the fixes | the layout of each record | records that cannot be read | none |
| 3.2 Choice of altitude channel | the GNSS altitude of the flight | the flight; fixes without a GNSS altitude | present on a share $\geq 0.95$ of the fixes, range $\geq 30\,\mathrm{m}$ |
| 3.3 Clock | the time of each fix | fixes that do not come after every fix before them | none |
| 3.4 Repeated positions | the position of each fix | fixes at the position of the fix before | none |
| 3.5 Fast tracks | the median step speed of the flight | the flight | $\tilde{v} \leq 70\,/\,110\,\mathrm{km/h}$ |
| 3.6 Horizontal excursions | the horizontal distance of each fix from the last fix accepted | fixes off the track; the flight is split where the track jumps | $v_{\max} = 100\,/\,175\,\mathrm{km/h}$, $W = 20\,\mathrm{s}$ |
| 3.7 Vertical excursions | the vertical distance of each fix from the last fix accepted | fixes off the track; the flight is split where the track jumps | $v_{z,\max} = 30\,/\,20\,\mathrm{m/s}$, $W$ |

unreadable altitude, including the zero the format uses for a missing reading [1], is marked as missing, and the fix keeps its time and position. A fix without a GNSS altitude is deleted later, by the altitude channel (Section 3.2).

Time is measured in seconds from the first fix. Records carry only the time of day, so a flight crossing UTC midnight would jump back by a day; a day is added when the clock passes from the last hour of one day to the first hour of the next. Other backward steps and repeated times are left to the clock rule of Section 3.3. Well-formed but implausible fixes, such as position spikes, are kept as well: the parser discards only what cannot be read. The validity flag, which marks fixes without a three-dimensional GNSS solution, is kept but not used at this stage.

## 3.2 Choice of altitude channel

We prefere to use GNSS altitude over the barometric one because the former is a more reliable absolute measure of the altitude above the seal level (ASL; Section 3.8 checks its datum). So, for every flight, $z = h^{\mathrm{GNSS}}$.

Let $N$ be the number of fixes read from the file, before any is deleted, and $P$ the set of those with a GNSS altitude. The flight is kept only if

$$\frac{|P|}{N} \geq 0.95 \qquad \text{and} \qquad \max_{i \in P} h_i^{\mathrm{GNSS}} - \min_{i \in P} h_i^{\mathrm{GNSS}} \geq 30\,\mathrm{m}. \tag{3.1}$$

The first condition rejects incomplete channels; the second rejects channels stuck at one value, and flights with almost no vertical excursion. Fixes flagged as lacking a three-dimensional solution count as present if they carry an altitude. Pressure altitude is never substituted for a missing GNSS value. In a flight kept, the fixes without a GNSS altitude are deleted, so every later step sees a time, a position and an altitude on every fix; by the first condition, this deletes at most a share $1 - 0.95$ of the fixes of a flight.

The barometer is checked with the same two conditions on $h^{\text{baro}}$, with thresholds 0.95 and 30 m, and with a third: its range must agree with the GNSS range,

$$\left|R^{\text{baro}} - R^{\text{GNSS}}\right| \le 0.1\, R^{\text{GNSS}}, \tag{3.2}$$

where $R$ is the highest minus the lowest altitude of a channel. Pressure altitude departs from true altitude by a few percent when the temperature departs from the standard atmosphere, so a barometer that measures altitude passes; some recorders write a raw pressure reading or feet in this field, and fail. A flight whose barometer passes the three conditions is a witness: its pressure altitude, measured independently of the GNSS, serves only to choose and to check the top vertical speed of the vertical excursion rule (Section 3.7), and never decides whether the flight is kept.

## 3.3 Clock

A recorder can write a time equal to or earlier than the one before it. On the fixes left by the altitude channel, in order, the clock rule keeps the first fix, and fix $i$ only if its time is later than that of every fix before it,

$$t_i > \max_{j<i} t_j. \tag{3.3}$$

Of several fixes with the same second, the first is kept. A backward step is deleted, and so is every fix after it until the clock passes the latest time already kept. The times left are strictly increasing, so every step between consecutive fixes has a positive duration (Section 3.5).

## 3.4 Repeated positions

Some recorders repeat their last position while the clock runs, so a fix can carry the coordinates of the one before it without being a new measurement. On the fixes left by the clock rule, numbered again in order, fix $i$ is deleted if

$$\varphi_i = \varphi_{i-1} \qquad \text{and} \qquad \lambda_i = \lambda_{i-1}. \tag{3.4}$$

The comparison is exact: the two records carry the same digits. Altitudes are not compared. In a run of fixes at the same position only the first is kept.

## 3.5 Fast tracks

A free flight spends part of its time climbing in thermals at a low ground speed, so the typical speed of its track stays well below the top speed of the wing. A track with a higher typical speed comes from something else, such as a powered aircraft or a file with compressed times, and the whole flight is dropped.

Let $n$ be the number of fixes left by the previous steps. Between fix $i$ and fix $i+1$ the horizontal distance is taken on a sphere of radius $R = 6{,}371$ km, with the haversine formula

$$d_i = 2R \arcsin \sqrt{\sin^2 \frac{\varphi_{i+1} - \varphi_i}{2} + \cos\varphi_i \cos\varphi_{i+1} \sin^2 \frac{\lambda_{i+1} - \lambda_i}{2}}, \tag{3.5}$$

with the angles in radians, and the step speed is

$$v_i = \frac{d_i}{t_{i+1} - t_i}, \qquad i = 1, \ldots, n-1. \tag{3.6}$$

The clock rule makes every denominator positive. The typical speed of the flight is the median of its step speeds,

$$\tilde{v} = \text{median}\{v_1, \ldots, v_{n-1}\}, \tag{3.7}$$

and the flight is kept if $\tilde{v}$ does not exceed 70 km/h for paragliders and 110 km/h for hang gliders. The rule deletes no fix, and the median is not moved by a few wrong steps, such as a position spike: those are left to the horizontal excursion rule (Section 3.6).

The thresholds were read off the distribution of $\tilde{v}$ over all flights (Figure 3.1). In both disciplines the bulk of the distribution ends below the threshold, and beyond it only a sparse tail of single flights remains. Split by wing class (Figures 3.2 and 3.3), the bulk of every class stays below the threshold. The faster shoulder of the hang-glider distribution comes from the rigid Class 2 wings.

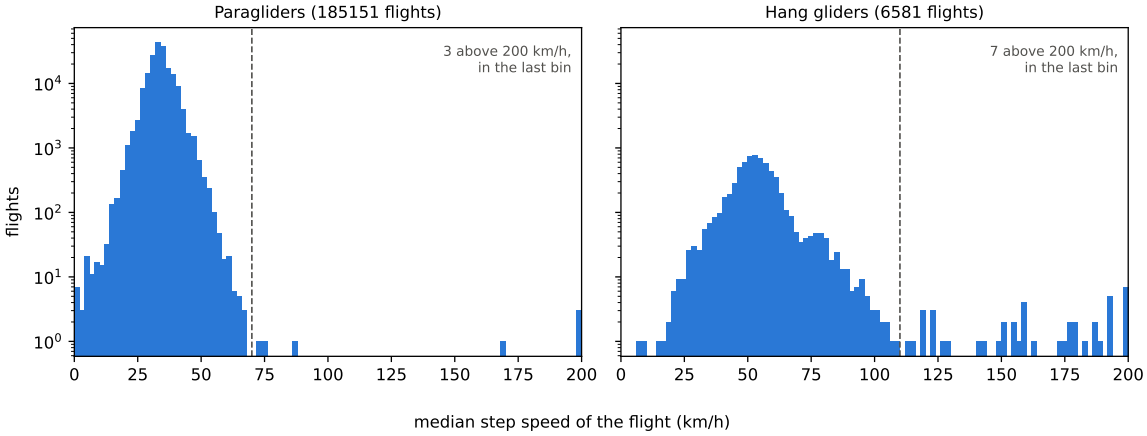



**Figure 3.1.** Median step speed $\tilde{v}$ (Equation (3.7)) of every flight past the altitude channel, computed on the fixes left by the previous steps. Bins of 2 km/h, logarithmic vertical axis; flights above 200 km/h fall in the last bin. Dashed line: threshold, 70 km/h for paragliders and 110 km/h for hang gliders.

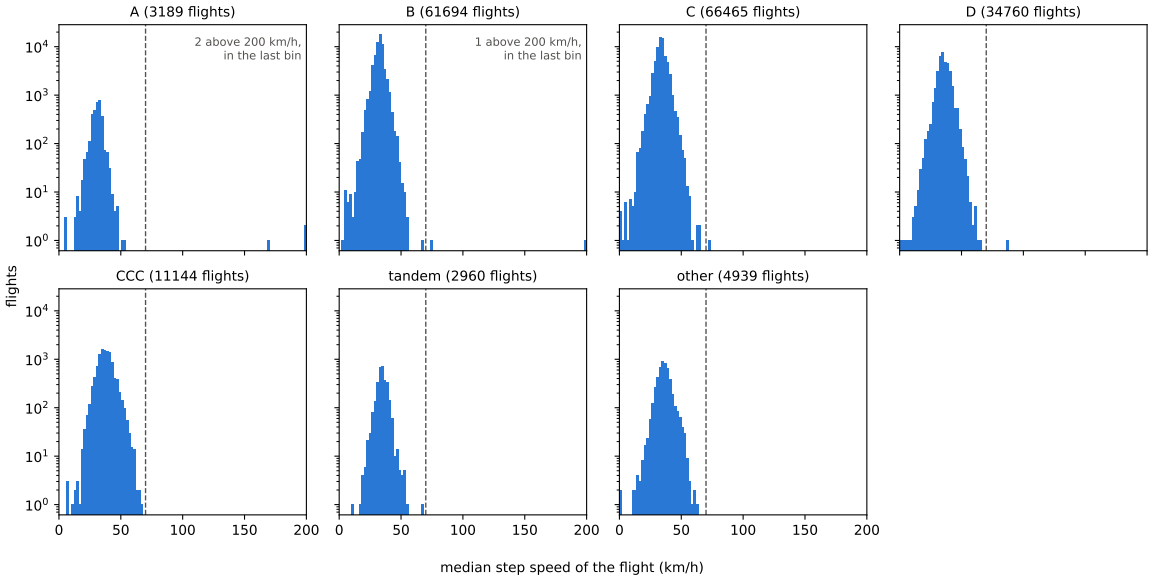



**Figure 3.2.** As Figure 3.1, for paragliders by wing class (Section 2.3): EN classes A to D, the CIVL Competition Class (CCC), tandem wings, certified or not, and every other label.

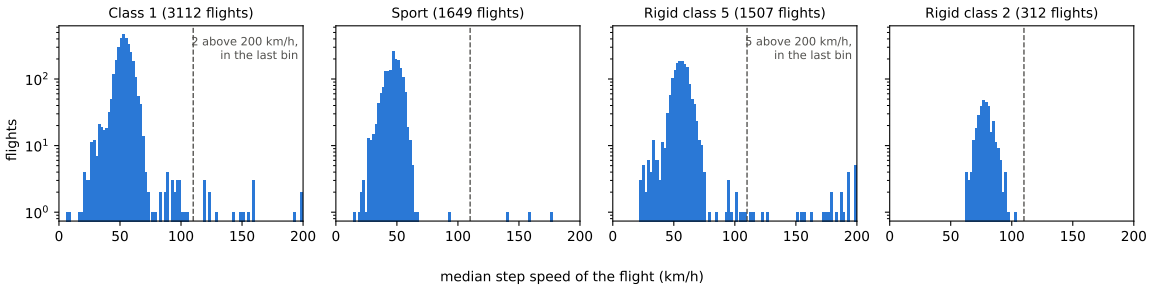



**Figure 3.3.** As Figure 3.1, for hang gliders by wing class (Section 2.3): Class 1, its Sport sub-class, and the rigid Classes 5 and 2. Flights of any other class are not shown.

## 3.6 Horizontal excursions

A step whose speed $v_i$, Equation (3.6), exceeds what any glider of the discipline can fly is a recorder error, but the step alone does not say what went wrong: the track may leave its path and come back, as in a spike, or go on from the new position, as in a jump. The excursion rule tells the two apart by looking ahead. It has two parameters: the top speed $v_{\max}$, read off the data, and a time $W$, set by hand.

The rule scans the fixes left by the previous steps in order, against an anchor $a$, the last fix accepted. Fix $j$ is reachable from the anchor if

$$d(a, j) \leq v_{\max}\,(t_j - t_a), \tag{3.8}$$

with $d(a, j)$ the distance of Equation (3.5). A reachable fix is accepted and becomes the anchor. An unreachable fix $j$ opens a block, and the rule looks for a later fix $k$ reachable from the anchor with $t_k - t_j \leq W = 20\,\text{s}$. If there is one, the track came back: fixes $j$ to $k-1$ are deleted and $k$ becomes the anchor. If there is none, the flight is split before $j$ and a new segment starts from $j$ (Figure 3.4). A block still open at the end of the track is deleted. The splits bound the segments of the flight, inside which the vertical rule runs (Section 3.7).

The first anchor of each segment, at the start of the track or after a split, is the first fix whose step to the next satisfies Equation (3.8), and the fixes before it are deleted. An isolated wrong fix disagrees with both its neighbours, so it never becomes an anchor, which would make the good fixes after it unreachable. The price is that a good fix followed by a spike is deleted with it.

The top speed must lie above the speed of real flight, and the rule tells which speed matters. Over a window of length $W$, the speed from fix $i$ is

$$u_i = \frac{d(i, \ell)}{t_\ell - t_i}, \qquad \ell = \min\{k :\ t_k - t_i \geq W\}. \tag{3.9}$$

If a stretch of real flight is faster than $v_{\max}$, its first fast step opens a block after an anchor $a$ on the track. The window from $a$ covers real flight and, unless it spans a gap, ends within $W$ of the opening, at a fix that is reachable if $u_a \leq v_{\max}$. A top speed above every window speed of real flight therefore never splits real flight: a fast stretch can lose its fixes, but the rule finds a reachable fix in time (Figure 3.5).

Computed from every fix, the window speed also follows real flight much further than the step speed: a fix displaced by a distance $D$ changes $u_i$ by at most $D/W$, against $D/\Delta t$ for each of the two steps of duration $\Delta t$ that touch it. Both distributions fall steeply (Figure 3.6, lower row) and then turn into a tail of errors, the step speed early, the window speed later and much lower. That tail holds the errors that last longer than $W$, such as a jump that

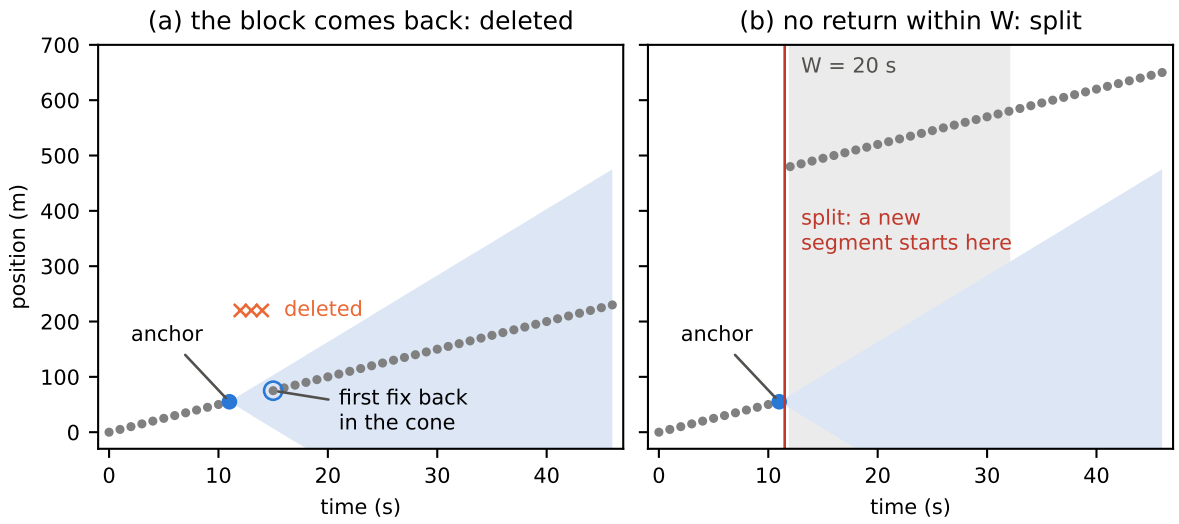



**Figure 3.4.** The excursion rule on two synthetic tracks along one coordinate, one fix per second, with a top speed lowered for legibility. Grey dots: fixes kept; blue dot: the anchor; shaded cone: the positions reachable from it, $|x - x_a| \le v_{\max}(t - t_a)$. (a) Three fixes leave the cone and the next comes back within $W$: the three are deleted (crosses). (b) The track jumps and stays: no fix comes back within $W = 20\,\mathrm{s}$ of the opening of the block (grey band), so the flight is split (vertical line) and a new segment starts from the block.

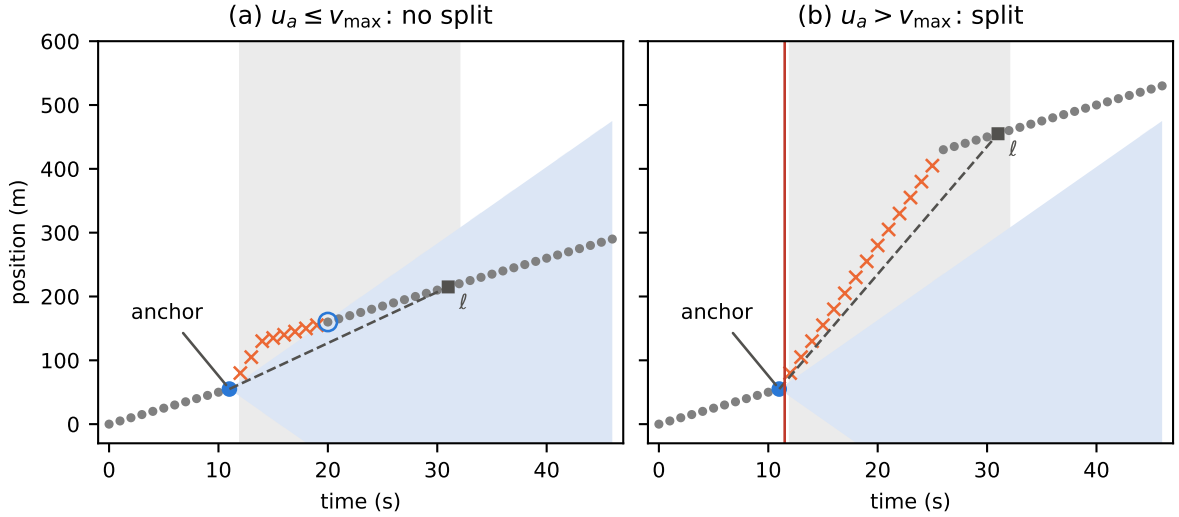



**Figure 3.5.** Real flight faster than the top speed, on synthetic tracks drawn as in Figure 3.4: a burst of fast steps opens a block after the anchor $a$, and the rule waits $W$ (grey band). Dashed line: from $a$ to $\ell$, the end of the window from $a$, Equation (3.9); its slope is $u_a$. (a) A short burst, $u_a \le v_{\max}$: $\ell$ lies in the cone and within the band, so the rule finds a reachable fix (circle), $\ell$ at the latest, and deletes only the fixes before it (crosses). (b) A burst that lasts most of the window, $u_a > v_{\max}$: no fix is reachable within $W$, and the rule splits real flight (vertical line); the fast fixes are deleted too, since the new segment starts at the first fix whose step to the next is reachable.

never comes back, and does not depend on the wing, while real flight does: split by wing class (upper row), the classes fall off in the order of their performance, then merge into one common tail. Real flight therefore ends where the fastest class reaches that tail. We set $v_{\max}$ there: 100 km/h for paragliders and 175 km/h for hang gliders.

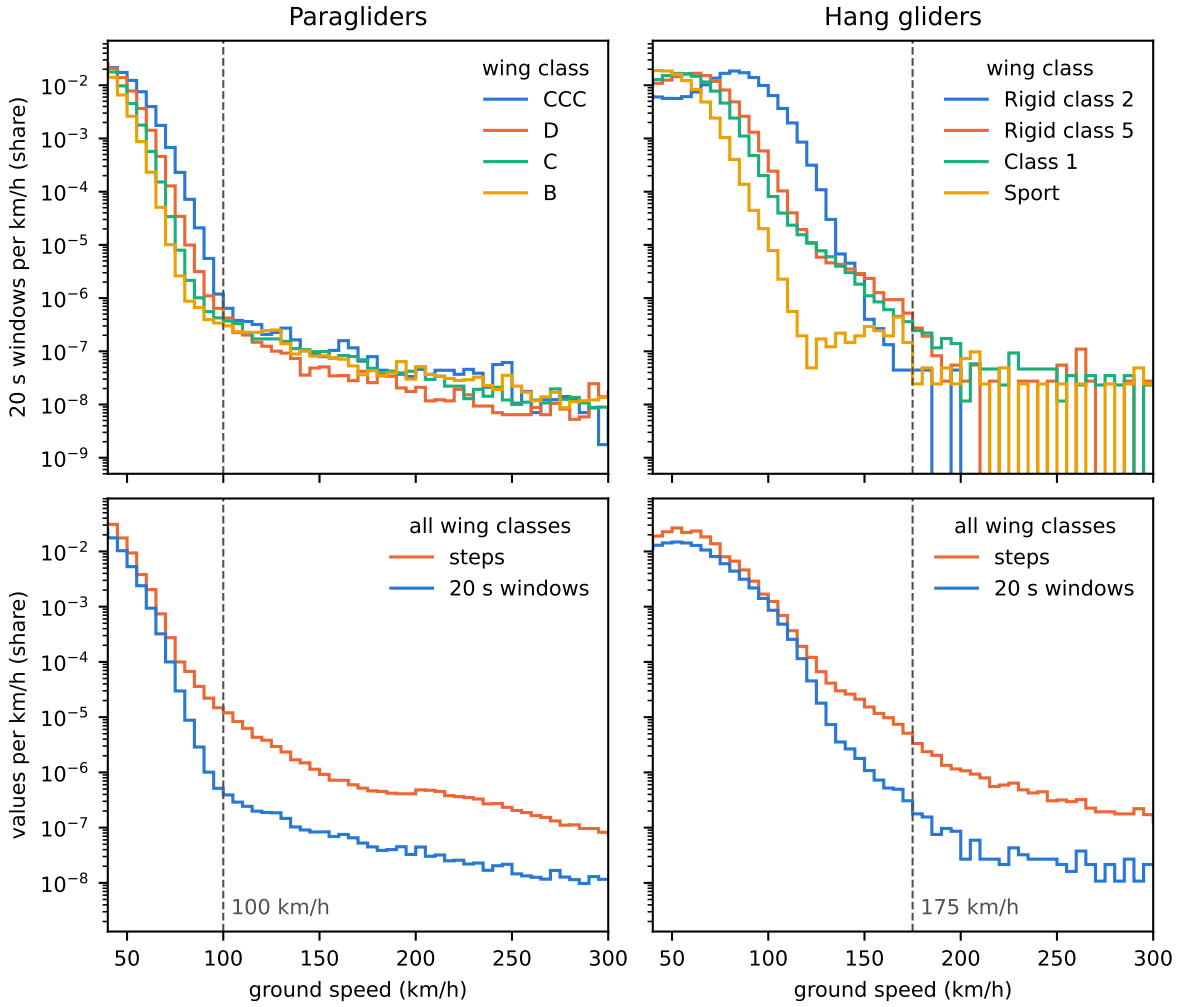



**Figure 3.6.** Ground speed in the flights kept by the previous steps. Upper row: speed over windows of length $W = 20$ s, Equation (3.9), for four wing classes of each discipline (Section 2.3). Lower row, all classes together: speed of single steps, Equation (3.6), and over windows. Each curve is the share of its values per km/h, in bins of 5 km/h, on a logarithmic axis; values beyond the axis count in the share. Dashed line: the top speed $v_{\max}$, 100 km/h for paragliders and 175 km/h for hang gliders.

With $W$ halved and doubled, 10 s and 40 s, and $v_{\max}$ unchanged (Table 3.2), the share of fixes deleted barely changes: almost every excursion that comes back does so within the shortest window. Every excursion that outlasts $W$ splits the flight, so the share of flights split roughly doubles when $W$ is halved and halves when it is doubled, and stays small for every window. A longer $W$ also bridges larger jumps: after a jump shorter than about $v_{\max}W$ the track counts as returned, and the fixes in between are deleted. The choice affects few flights, and we keep the middle value, which still leaves the slowest common recorders, sampling every 10 s (Figure 2.1), two fixes to come back.

**Table 3.2.** The excursion rule with the window $W = 20\,\mathrm{s}$ and with half and twice $W$, in the flights kept by the previous steps. Flights split: share of the flights split at least once. Fixes deleted: share of the fixes the rule receives that it deletes.

| | Paragliders (185,321 flights) | | Hang gliders (6,536 flights) | |
|---|---|---|---|---|
| $W$ (s) | Flights split (%) | Fixes deleted (%) | Flights split (%) | Fixes deleted (%) |
| 10 | 1.99 | 0.031 | 1.18 | 0.014 |
| 20 | 0.90 | 0.033 | 0.67 | 0.014 |
| 40 | 0.42 | 0.034 | 0.32 | 0.015 |

## 3.7 Vertical excursions

The GNSS altitude has errors of its own: it can spike or jump while the position follows the track. The horizontal rule cannot see them, since it judges positions only, so the same rule runs again on the altitude $z$, inside each segment the horizontal rule left: across a split, the two altitudes belong to different pieces of the track. In analogy with Equation (3.8), fix $j$ is reachable from the anchor $a$ if

$$|z_j - z_a| \le v_{z,\max}\,(t_j - t_a), \tag{3.10}$$

with $v_{z,\max}$ the top vertical speed of the discipline. Blocks, splits and first anchors work as in Section 3.6, with the same window $W$: the reason for keeping it, two fixes for the slowest recorders to come back, holds for any coordinate. The new splits add to the horizontal ones, and the rule deletes whole fixes, positions included. This is a choice: every fix left then carries a time, a position and an altitude, as after the altitude channel (Section 3.2), and the fixes deleted are few (Table 3.3).

As for $v_{\max}$, a top vertical speed above every vertical speed of real flight over a window of length $W$ never splits a real descent, however steep. In analogy with Equations (3.6) and (3.9), the vertical speeds of a step and over a window are

$$w_i = \frac{|z_{i+1} - z_i|}{t_{i+1} - t_i}, \qquad \bar{w}_i = \frac{|z_\ell - z_i|}{t_\ell - t_i}, \tag{3.11}$$

with $\ell$ as in Equation (3.9), both inside one segment. To read $v_{\max}$ we had the wing classes; for $v_{z,\max}$ we have more, since in most flights the barometer measures the same motion independently of the GNSS. In the flights whose barometer is a witness (Section 3.2), its speeds, Equation (3.11) with $h^{\mathrm{baro}}$ in place of $z$, show real vertical motion without the errors of the GNSS (Figure 3.7).

Over windows the barometer falls steeply, alike for every wing class (upper row), then flattens into a floor. In the lower row the GNSS lies above the barometer while both fall, since it adds its own errors, and beyond the fall the two merge into one floor: speeds no glider sustains over $W$, shared by both channels, hence errors common to the two. Real vertical motion therefore ends where the fall of the barometer meets the floor, and we set $v_{z,\max}$ there: 30 m/s for paragliders and 20 m/s for hang gliders, whose fewer flights make the curves noisier and the reading coarser. The same threshold judges the flights without a witness: the vertical speed of real flight depends on the wing and on the air, which the recorder does not change.

The barometer also checks the threshold. On the pressure altitude of the witness flights the rule deletes far fewer fixes and splits fewer flights than on the GNSS altitude (Table 3.3). There it can only remove real flight or an error the two channels share, so this bounds from above the real flight it removes from the GNSS altitude.

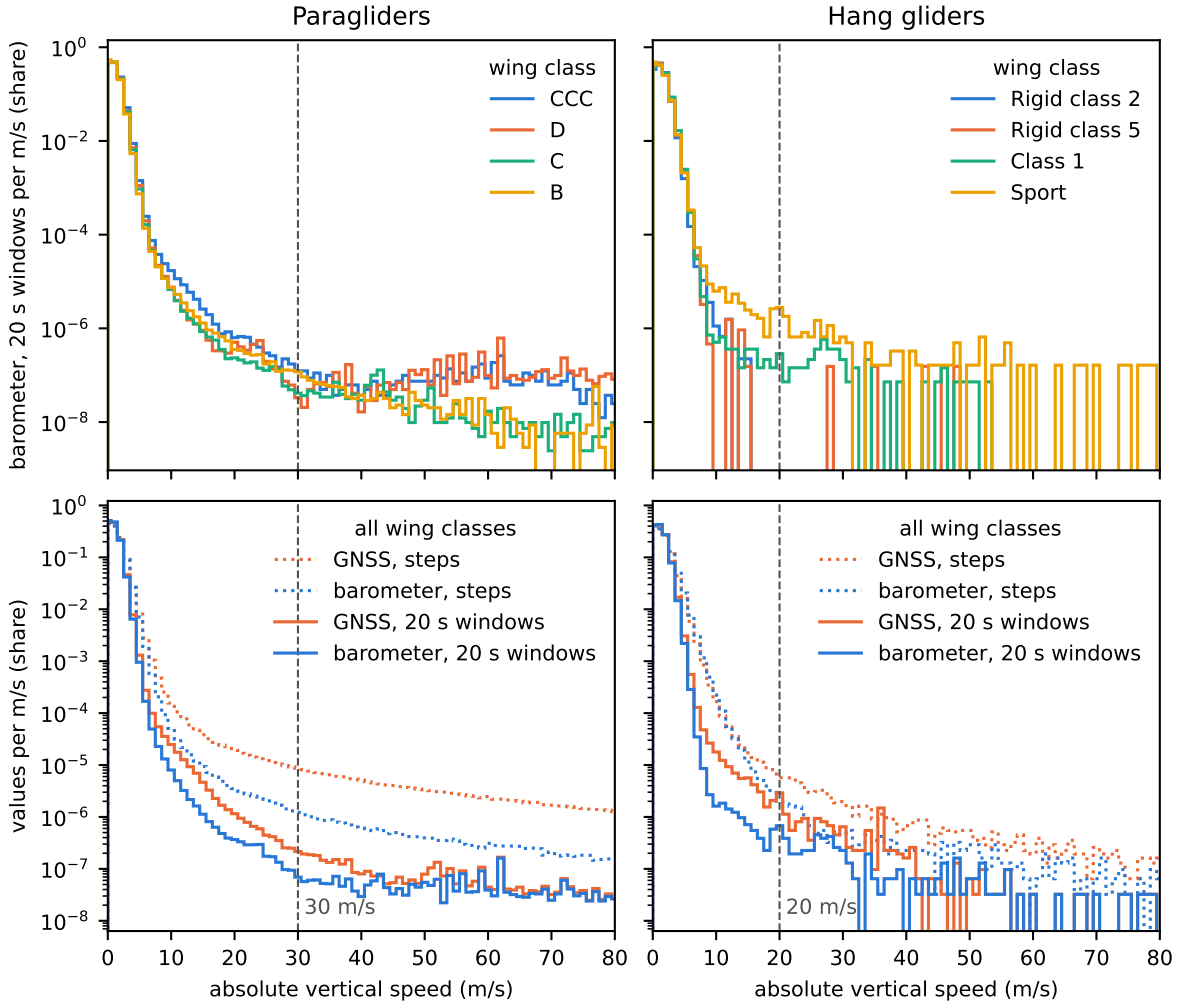



**Figure 3.7.** Absolute vertical speed in the flights kept by the previous steps whose barometer is a witness (Section 3.2), on the steps and windows where both altitudes are known. Upper row: speed of the pressure altitude over windows of length $W = 20\,\mathrm{s}$, $\bar{w}_i$ of Equation (3.11), for four wing classes of each discipline (Section 2.3). Lower row, all classes together: speed of the GNSS and of the pressure altitude over single steps, $w_i$, and over windows. Each curve is the share of its values per m/s, in bins of 1 m/s centred on whole values (the first, from 0, half as wide), on a logarithmic axis; values beyond the axis count in the share. Dashed line: the top vertical speed $v_{z,\max}$, 30 m/s for paragliders and 20 m/s for hang gliders.

**Table 3.3.** The vertical excursion rule with the top vertical speed $v_{z,\max}$ and the window $W = 20\,\mathrm{s}$, in the flights kept by the previous steps: on the GNSS altitude of every flight, and on the pressure altitude of the flights whose barometer is a witness (Section 3.2). Flights split: share of the flights split at least once. Fixes deleted: share of the fixes the rule receives that it deletes.

| | Altitude | Flights | Flights split (%) | Fixes deleted (%) |
|---|---|---|---|---|
| Paragliders | GNSS | 185,321 | 0.41 | 0.047 |
| | pressure | 125,856 | 0.08 | 0.003 |
| Hang gliders | GNSS | 6,536 | 1.12 | 0.047 |
| | pressure | 5,393 | 0.32 | 0.002 |

## 3.8 Local coordinates and vertical datum

The analysis works in local Cartesian coordinates: for every flight, $x$ east and $y$ north of its first fix and $z$ the altitude, all in metres. This section fixes the reference surfaces behind them and checks, flight by flight, whether the GNSS altitude is a height above sea level.

Latitude $\varphi$ and longitude $\lambda$ are geodetic coordinates on the WGS84 ellipsoid: both specifications require it [1, 5], and every file that declares a horizontal datum declares WGS84. The datum of the GNSS altitude varies: the IGC specification prescribes the height above the WGS84 ellipsoid [1], the CIVL specification for hang gliders and paragliders the height above the geoid [5]. The height above the ellipsoid, $h$, and the height above the geoid, $H$, which lies close to mean sea level, differ by the geoid undulation $N$:

$$h = H + N(\varphi, \lambda), \tag{3.12}$$

with $N$ between 43 and 57 m over metropolitan France in the EGM2008 model [6]. A file can declare its datum in the header `HFALG`, as `GEO` or `ELL` [1], but 73 % of the tracks declare nothing, and some declarations are false: XC Tracer firmware versions that declare `ELL` record $H$, and XCTrack on some phones declares `GEO` and records $h$. The data show the datum actually used. Flights that start from the same place stand on the same ground, so their first altitudes should agree. We compared each flight with the flights of one large group, XCTrack with `GEO`, started within 25 m of it; this is possible for 52 % of the flights. Of these, 90 % agree with the group, 2.4 % are higher by $N$, and up to 3 % are lower by $N$, as if the correction had been applied with the wrong sign.

We place positions on the ellipsoid and heights on the geoid, as topographic maps do.

- **Horizontal.** Each fix is moved to the point of the ellipsoid below it, $h = 0$, converted to Earth-centred Cartesian coordinates,

  $$(X, Y, Z) = (\nu \cos\varphi \cos\lambda,\ \nu \cos\varphi \sin\lambda,\ \nu(1-e^2)\sin\varphi), \qquad \nu = \frac{a}{\sqrt{1 - e^2 \sin^2\varphi}}, \tag{3.13}$$

  with $a$ and $b$ the semi-axes of the WGS84 ellipsoid [1] and $e^2 = 1 - b^2/a^2$, and rotated into the east and north directions of the first fix $(\varphi_0, \lambda_0)$:

  $$x = -\sin\lambda_0\, \Delta X + \cos\lambda_0\, \Delta Y, \qquad y = -\sin\varphi_0 \cos\lambda_0\, \Delta X - \sin\varphi_0 \sin\lambda_0\, \Delta Y + \cos\varphi_0\, \Delta Z, \tag{3.14}$$

  where $\Delta X = X - X_0$, and likewise for $Y$ and $Z$, with $(X_0, Y_0, Z_0)$ the first fix. The ellipsoid is the surface on which $\varphi$ and $\lambda$ are defined; a sphere of radius $R = 6{,}371$ km would stretch east–west and north–south distances differently, by about 0.3 % over France. With $h = 0$, $x$ and $y$ depend on latitude and longitude only, so the datum of the altitude never enters them. With the recorded altitude as $h$, a fix at distance $d$ from the origin would also move outwards by $zd/R$: 31 m for $z = 2{,}000$ m at $d = 100$ km.

- **Vertical.** $z$ is the recorded GNSS altitude, a height above the geoid in the flights that pass the check below. The terrain model, BD ALTI, IGN's RGE ALTI on a 25 m grid, gives the normal altitude of the ground, a height above sea level in the French levelling systems: IGN69 on the mainland, IGN78 in Corsica [7]. In those flights the height above the ground is $z - z_\mathrm{t}(\varphi, \lambda)$, with the terrain read at the latitude and longitude that give $x$ and $y$. The third component of the rotation, the height above the plane tangent at the first fix, is not used: it falls below the altitude by about $d^2/2R$, 785 m at $d = 100$ km.

The check is made on the ground. In the stretch after the landing in which the glider stands still, its altitude should be that of the terrain. With $\Delta$ the median of $z - z_\mathrm{t}$ over this

stretch, a flight gets the height above the ground if

$$|\Delta| \leq w = 20\,\mathrm{m}. \tag{3.15}$$

The tolerance lies between the scatter of the GNSS altitude, since at a shared takeoff three flights in four start within 10 m of the median altitude, and $N$: a file that records $h$, with $\Delta \approx N$, fails the check, and so does a flight without a still stretch after the landing. Geoid models and the zero of the French levelling network differ by much less than $w$: over metropolitan France, EGM96 and EGM2008 by at most 2.3 m, NGF-IGN69 and EGM2008 by 0.1 to 1.1 m. BD ALTI is accurate to 0.2–0.7 m where it comes from LiDAR or aerial photographs, and to about 7 m on the steep slopes of mountain areas [7]. The landing is used because landing fields are flat: a few metres of position error barely change the altitude read from the grid.

BD ALTI covers metropolitan and overseas France only [7]: a flight that lands outside it cannot be checked, and a fix outside it has no terrain altitude. 3.4 % of the paraglider flights and 4.7 % of the hang-glider flights start or end outside it. No flight is dropped: every flight keeps $x$, $y$ and $z$ and stays in the analyses built on positions and altitude increments, since within a flight the datum does not change and $N$ changes by at most 21 cm per kilometre. The analyses of the height above the ground, or of the altitude above sea level, use only the flights that pass Equation (3.15).

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